Week of 31st

Monday, 30 October 2023 4:51 PM

## Realistic, & practical flow:

Turbulent flow: can use the same equations (microscopic balances)

↳ However, flow is no longer unidirectional & SS } time dependent and 3D

$v_\theta, v_r \neq 0$

↓

Cannot solve for velocity profile!

↓

Use experiments to fill in the gaps! → Empirical correlations

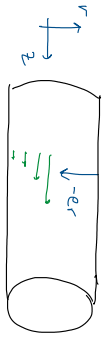
Pressure drop vs. flow rate relationships ($F_{drag}$ vs. $Q$)

Fluid force on a surface S: $\underline{F} = \iint_S [\hat{n} \cdot \underline{\underline{\Pi}}]\big|_{@surface} \, dS$

This force is being exerted by the fluid on the surface

↓

This is the cause of pressure drop in pipe flow!!



$\underline{F}$ ⇒ total force

$\underline{F}_{drag} = \underline{F}_z$ (i.e. $z^{th}$ component of force)

$\therefore \; \underline{F}_{drag} = \int_0^L \int_0^{2\pi} \hat{e}_z \cdot (-\hat{e}_r \cdot \underline{\underline{\Pi}}\big|_R) \, R \, d\theta \, dz$

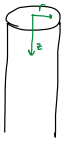
↓ non-dimensionalize

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu \left( \frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z} \right)\Big|_R R \, d\theta \, dz$$

Fluid drag in turbulent flow on a pipe surface of unit normal $-\hat{e}_r$

## Practice Questions

Laminar flow in a pipe: (Unidirectional, SS) } Continuity + Navier → BC → v profile



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{r\theta z}$ ; $\underline{g} = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}_{rθz}$ $\quad p = p(z)$

↓

+ Pressure profile: $p(z) = -\left(\frac{p_0 - p_L}{L}\right) z + p_0$

+ Velocity profile: $v_z(r) = \frac{(p_0 - p_L + \rho g L) R^2}{4\mu L} \left[ 1 - \left(\frac{r}{R}\right)^2 \right]$

$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$

$= -pI + \begin{pmatrix} 0 & 0 \\ 0 & \mu \frac{\partial v_z}{\partial r} \end{pmatrix}$

$\Delta \quad Q = \langle v \rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$

$$\underline{\underline{\Pi}}(r,z) = -p\underline{\underline{I}} + \underline{\underline{\tau}}$$

$$= -p\underline{\underline{I}} + \begin{pmatrix} 0 & 0 & \mu\frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \mu\frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix}_{rθz}$$

$$= \begin{pmatrix} -p(z) & 0 & (p_L - p_0)\frac{r}{2L} \\ 0 & -p(z) & 0 \\ (p_L - p_0)\frac{r}{2L} & 0 & -p(z) \end{pmatrix}$$

$$\Delta \quad Q = \langle v \rangle \pi R^2 = \frac{\pi (p_0 - p_L) R^4}{8\mu L}$$

$$\langle v \rangle = \frac{(p_0 - p_L) R^2}{8\mu L}$$

→ Neglect gravity:

$$v_z(r) = \frac{(p_0 - p_L + \rho g L)^{0} R^2}{4\mu L}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$= \frac{(p_0 - p_L) R^2}{4\mu L} \cdot \frac{2}{2}\left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$v_z(r) = 2\langle v \rangle \left[1 - \left(\frac{r}{R}\right)^2\right]$$

$$\frac{\partial v_z}{\partial r} = \frac{-4\langle v \rangle r}{R^2}$$

$$\underline{F}_{drag} = \int_0^L \int_0^{2\pi} -\mu\left(\frac{\partial v_z}{\partial r} + \frac{\partial v_r}{\partial z}^{0}\right)\Big|_R R\, d\theta\, dz$$

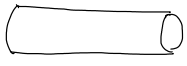
$$= \int_0^L \int_0^{2\pi} -\mu\left(\frac{-4\langle v \rangle r}{R^2}\right)\Big|_R R\, d\theta\, dz$$

$$= (4\mu\langle v \rangle)(2\pi \cdot L)$$

$$= \cancel{4}\cancel{\mu}\cancel{2\pi}\cancel{L} \cdot \left(\frac{(p_0 - p_L) R^2}{\cancel{8}\cancel{\mu}\cancel{L}}\right)$$

$$\boxed{\underline{F}_{drag} = \pi R^2 (p_0 - p_L)}$$

## Turbulent flow in smooth pipes:



Prandtl correlation: $\frac{1}{\sqrt{f}} = 4\log\left[Re\sqrt{f}\right] - 0.4$ (Smooth pipes)

7.156

$$Re = \frac{\rho\langle v \rangle D}{\mu}$$

$f = \frac{\Delta p D}{2L\rho\langle v \rangle^2}$ (Pipe friction factor from experiments)

### Algorithm 1:

1) Guess $Q_{guess}$ (will give $\langle v \rangle$)
2) Find $Re$
3) Find $f_1$ ← prandtl correlation
4) Find $\Delta p_1$ ← friction factor
5) Calculate $\Delta p_2 = \text{known value} - \Delta p_1$
6) Solve for "LHS" (given $\Delta p_2$) → gives $f_2$ → $\langle v \rangle$ (using friction factor)

$$\Rightarrow 4\log\left[\frac{\rho\cancel{\langle v \rangle} D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho\cancel{\langle v \rangle^2}}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\frac{\rho D}{\mu}\sqrt{\frac{\Delta p D}{2L\rho}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

$$\Rightarrow 4\log\left[\sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}}\right] - 0.4 = \frac{1}{\sqrt{f}}$$

5) Calculate $\Delta p_2$ = known value − $\Delta p_1$

6) Solve for "LHS" (given $\Delta p_2$) → gives $f_2$ → $\langle v \rangle$ (friction factor) using

7) Solve for $Q_{calc}$ [$Q_{calc} = \langle v \rangle \pi R^2$]

8] Compare this to step 1 $Q_{guess}$ → if close then stop
↳ else iterate

$$\left[ \frac{\mu}{L} \sqrt{\frac{2}{\rho \Delta p}} \right] \quad \text{if}$$

$$\Rightarrow \underbrace{4 \log \left[ \sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}} \right] - 0.4}_{\text{"LHS"}} = \frac{1}{\sqrt{f}}$$

## Turbulent flow in connected smooth pipes

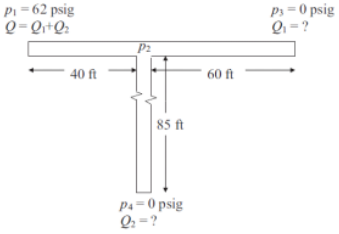
$$\Delta p_2 = P_2 - P_3 = P_2 - P_4$$

1) Guess $\Delta p_2$
↓ "LHS"
find $f$
↓ friction factor
$\langle v \rangle$
↓ ↘
$Q_1$ $Q_2$

2) Total $Q$ ($Q_1 + Q_2 = Q$)
↓
Now use Algorithm 1 !
(until step 5)
↳ $\Delta p_1 = p_1 - p_2$

3) $(\Delta p_2)_{calc} = 62 - \Delta p_1$

4) Iterate until $(\Delta p_2)_{calc} \approx (\Delta p_2)_{guess}$




**EXAMPLE 7.9.** *A 40-foot section of 1.0-inch ID piping branches into two pipes of the same diameter, one of which is 60.0 feet long and one of which is 85.0 feet long (Figure 7.23). The main pipe is connected to the municipal water supply, which supplies a constant water pressure of 62 psig at the pipe entrance. What are the flow rates through the two pipe exits? What is the pressure at the splitting point? Assume smooth pipes.*

① Since we don't know $Q$
↳ we first need to solve for $Q_1$ & $Q_2$ by guessing $\Delta p_2$ value

② Once we know $Q$ → then focus on small initial 40 ft pipe and calculate $\Delta p_2$ value independtly
↓
See if it matches?
↳ if not we iterate!

## Turbulent flow in connected rough pipes

Colebrook correlation: $$\frac{1}{\sqrt{f}} = -4 \log \left( \frac{\varepsilon}{D} + \frac{4.67}{Re\sqrt{f}} \right) + 2.28$$

$$\underbrace{\qquad\qquad}_{g(f)}$$

↓

$$f = \frac{\Delta p D}{2 L \rho \langle v \rangle^2} \quad ; \quad Re = \frac{\rho \langle v \rangle D}{\mu}$$

↓

$$\frac{1}{\sqrt{f}} = -4 \log \left( \frac{\varepsilon}{D} + \frac{4.67}{\left( \sqrt{\frac{\rho \Delta p D^3}{2L\mu^2}} \right)} \right) + 2.28$$

"LHS" (when we only know pressure drop & no average velocity)

**PROBLEM: 7.16**

For steady pipe flow, repeat the branched-piping example, Example 7.9, for pipes with roughness $\varepsilon = 0.005$.